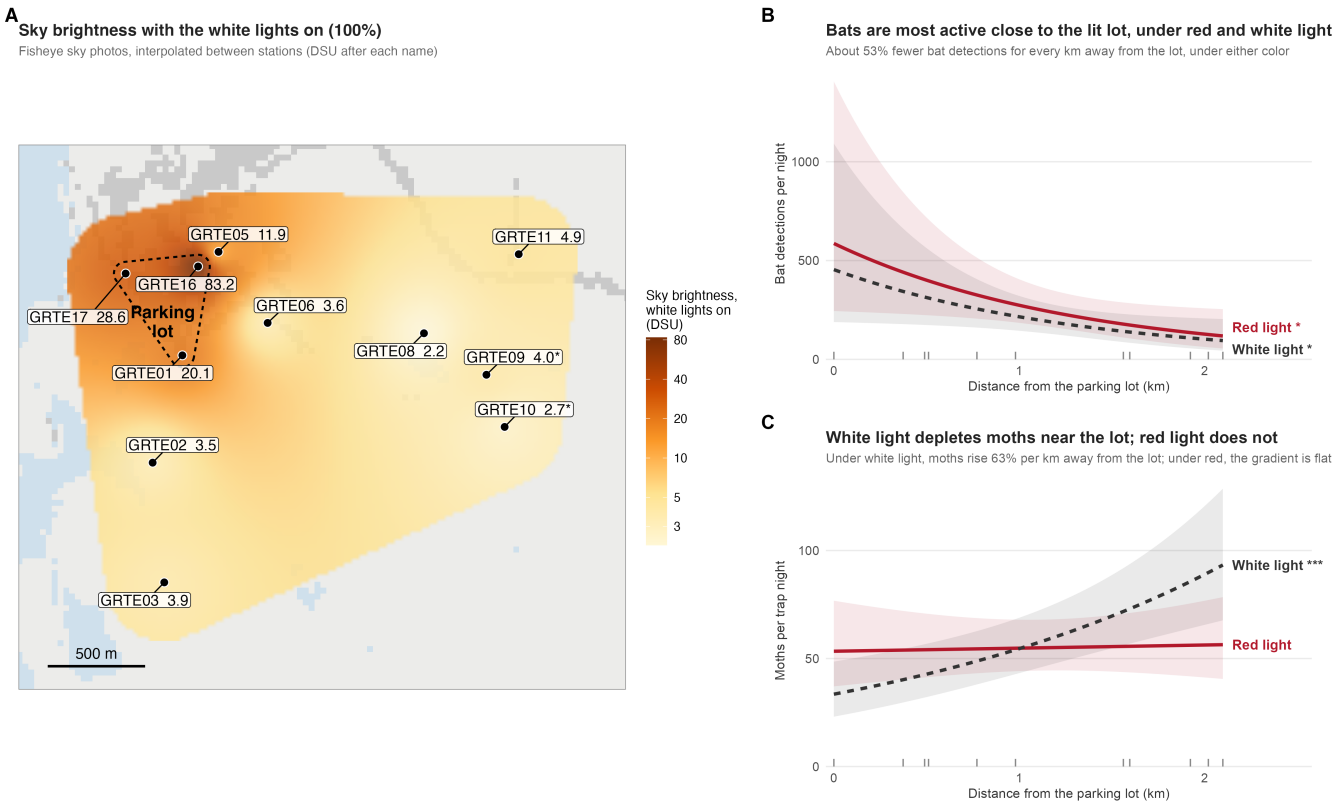


Figure 1: Sky brightness and the distance gradient at the lit Colter Bay parking lot (stations GRTE01, GRTE16, GRTE17), Grand Teton National Park, 2022; bats on acoustic detectors and moths in light traps while the lot's lights were switched night by night between red and white and among five intensities (10–100%). **B**–**C** use the 11 stations within 0–2.1 km of the lot (negative binomial GLMMs with light color × intensity, color × distance, moon phase, non-forest cover, sky brightness, season and a site random effect; light color and intensity were set experimentally, while the distance pattern is observational). **A** Sky brightness with the lot's white lights on at 100%: mean of each station's three fisheye sky photos (dark sky units, DSU; value after each name), interpolated between stations by inverse-distance weighting on log DSU and clipped to 300 m around them; dashed outline = parking lot (GRTE01, GRTE16, GRTE17); * GRTE09 and GRTE10 have no white-light photos, so their lights-off values are used (stations this far from the lot change little with the lights on). The far stations, not shown, measured 3.0 (GRTE13), 9.6 (GRTE14) and 3.6 DSU (GRTE15). **B** Predicted bat detections per night and **C** predicted moths per trap night along the gradient under red (solid) and white (dashed) light, ± 95% CI; ticks on the x-axis mark the stations. Asterisks on the line labels test that color's change with distance (* $p < 0.05$, *** $p < 0.001$).

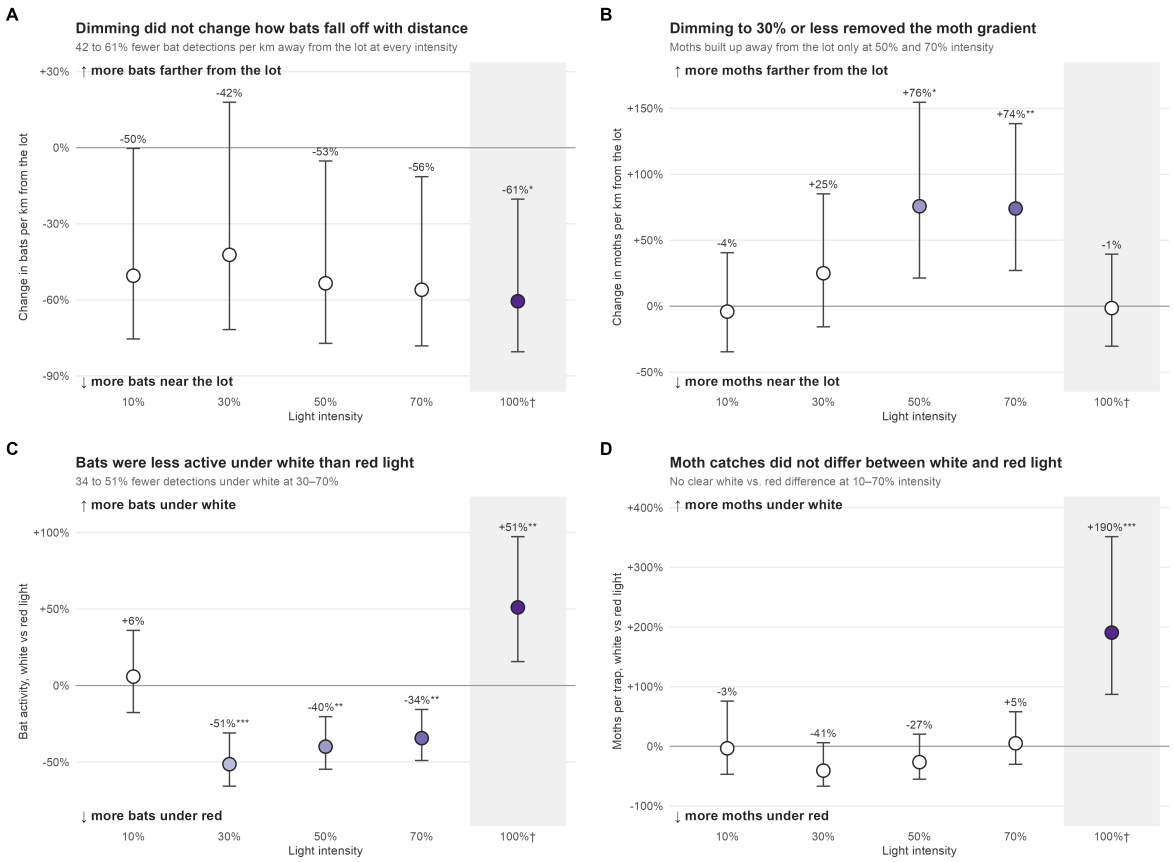


Figure 2: Distance and color effects at each light intensity, from the same models as Figure 1B–C. **A** Bats and **B** moths: % change in detections per km away from the lot at each intensity. **C** Bats and **D** moths: % difference in detections under white vs. red light at each intensity (at the mean distance). Points shaded from pale to deep purple for 10% to 100% intensity; filled points differ from zero (Holm-adjusted across the five intensities), open points do not; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; 95% CIs. † 100% (shaded): the white vs. red differences hold after accounting for moon, temperature and rain, but the red-light nights at 100% included the July full moon and the two wettest nights of the study (moths: four trap nights; bats: six recording nights), and the bat difference runs against the pattern at lower intensities.

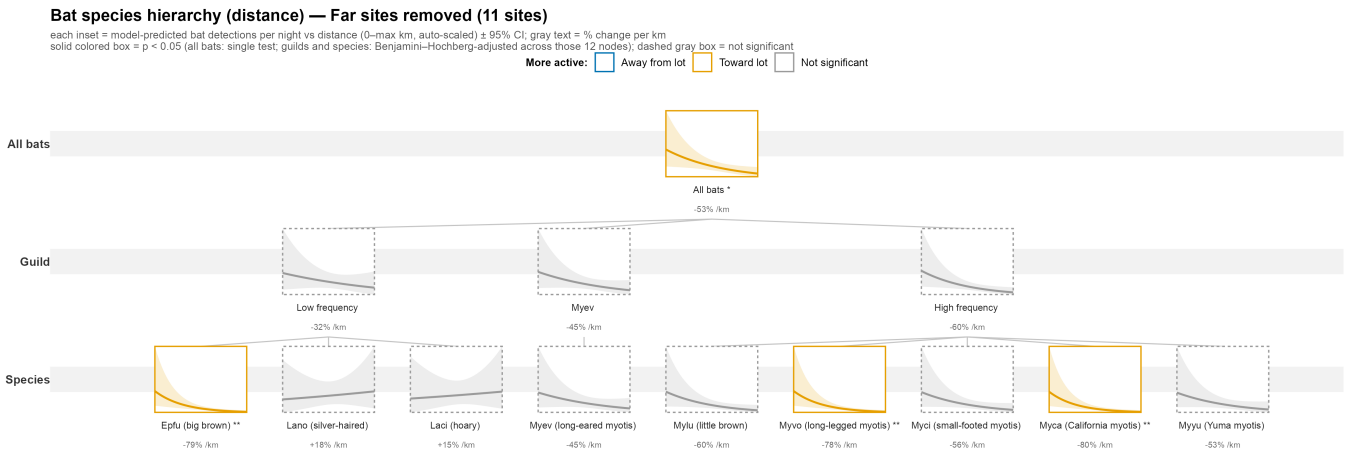


Figure 3: Bat distance responses by tier (all bats → call-frequency guild → species), 11 stations within 2.1 km of the lot. Each inset is model-predicted bat detections per night vs. distance, ± 95% CI (auto-scaled), drawn in its box's color; gray text under each node is its % change per km. Solid colored box = significant (all bats: single test; guilds and species: Benjamini–Hochberg-adjusted across the 12 nodes); orange = more active toward the lot; dashed gray = not significant (gray curve); * $p < 0.05$, ** $p < 0.01$. Zero-inflated negative binomial GLMMs of detections per site-night with light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random effect.

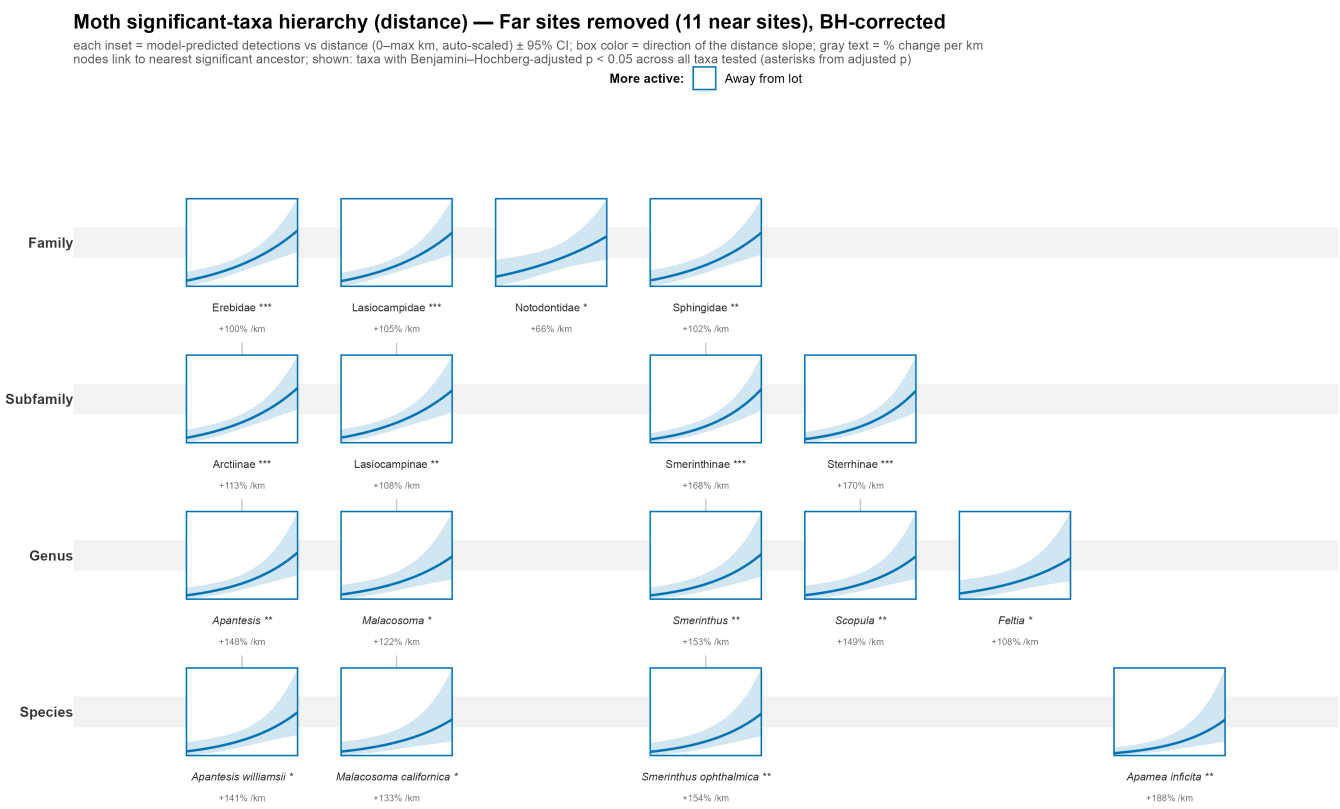


Figure 4: Moth taxa with a significant distance slope, 11 stations within 2.1 km of the lot: Benjamini–Hochberg-adjusted $p < 0.05$ across the 51 taxa tested (taxa with 30 site-nights), each linked to its nearest significant ancestor. Each inset is model-predicted moths vs. distance, ± 95% CI (auto-scaled), drawn in its box's color; gray text = % change per km; blue box = more active away from the lot; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (adjusted). Taxon × distance zero-inflated negative binomial GLMM with light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random effect.

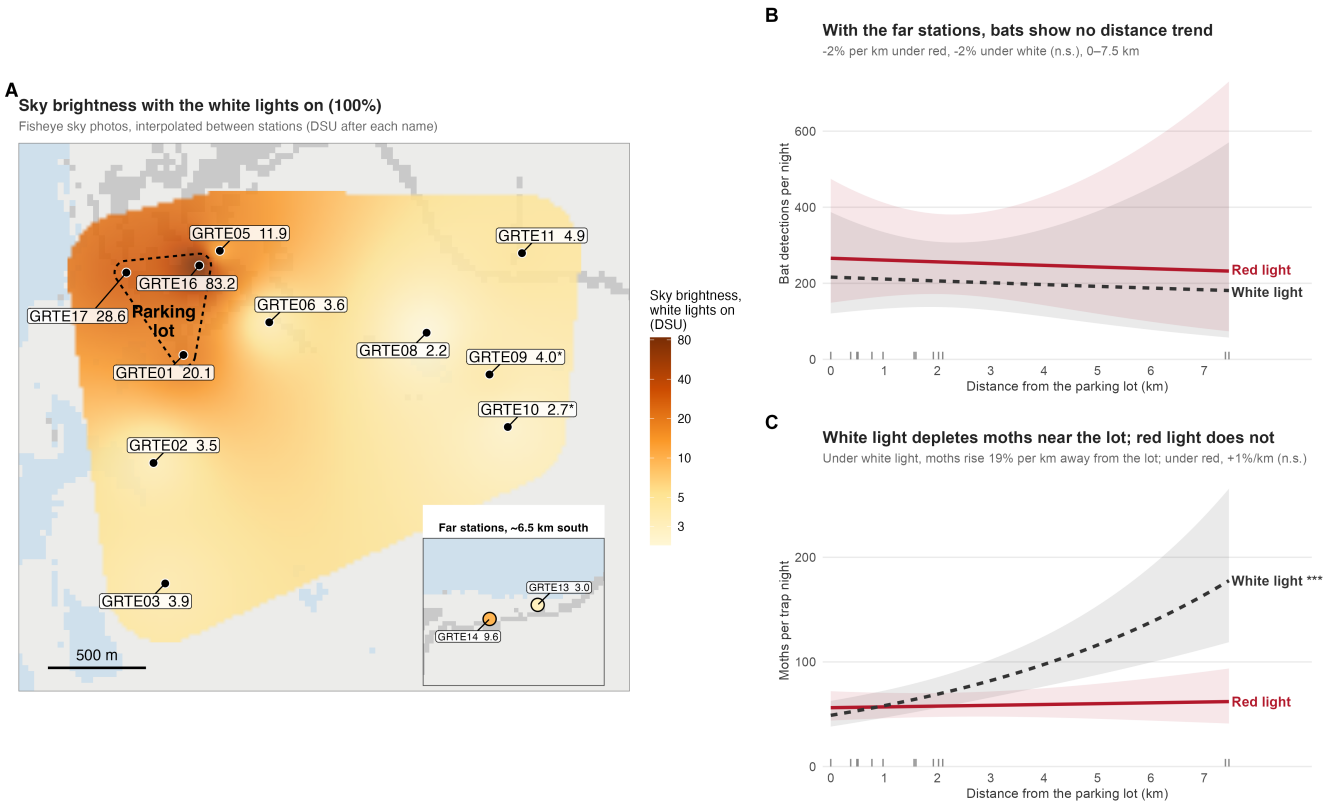


Figure 5: As Figure 1, but with the far stations included (13 stations, 0–7.5 km; GRTE15 excluded). **A** Sky brightness with the white lights on (as Figure 1A); inset: the two far stations, about 6.5 km south near Signal Mountain, colored on the same scale (not interpolated). **B** Predicted bat detections per night and **C** predicted moths per trap night along the full gradient under red (solid) and white (dashed) light, ± 95% CI; ticks on the x-axis mark the stations. Asterisks on the line labels test that color's change with distance (*** $p < 0.001$); n.s. = not significant.